Jonathan Barnes | Data Scientist

Minneapolis, Minnesota | Jonathanmbarnes@outlook.com | https://jonathanmbarnes.github.io/

PROFESSIONAL EXPERIENCE

University of Minnesota, Minneapolis, MN

Analytics Engineer | Center for Public Health Systems *(May 2025 - Present)*

- Designed and maintain an in-house data warehousing and analytics platform for federal public health funding analysis.
- Integrated and standardized 3M+ HHS award records across grants, contracts, prime awards, and sub-awards.
- Implemented normalization and award-recipient linkage logic across complex funding and organizational structures.
- Engineered a columnar storage architecture using Parquet and DuckDB, reducing data size by 75%+.
- Developed an interactive Tableau dashboard undergoing refinement for public release and granular funding analysis.
- Analyzed PHIG grant spending and public health systems for grant-funded evaluation.

Graduate Teaching Assistant - In-Person and Virtual | Division of Biostatistics and Health Data Science *(Aug 2024 - May 2025)*

- Taught weekly lab sessions in R and SAS to develop student proficiency in statistical analysis and programming.
- Evaluated student projects and quizzes, providing targeted feedback to improve statistical reasoning and code quality.

Graduate Research Assistant | Medical School *(Oct 2023 - May 2024)*

- Classified plasma biomarker profiles from NIA funded study into cutoff-based phenotypes, driving downstream modeling.
- Modeled time to cognitive decline with Cox proportional hazards regression, assessing predictive power for dementia.
- Programmed differential gene expression analysis stratified by grouping and recorded cognitive decline.

University of St. Thomas, St. Paul and Minneapolis, MN

ITS Support Technician Intern & Senior Student Technician | ITS Support Services *(Aug 2021 - Aug 2023)*

- Resolved or consulted on at least 4,500 departmental tickets over two years.
- Engineered and developed a Python application vital to ITS service desk function and operational efficiency.
- Mentored junior student technicians to independently manage advanced service desk operations and faculty consultations.
- Imaged and administered software on over 1,500 summer replacement computers, over double the usual volume.
- Assembled in-house virtual lab to reduce existing AWS EC2 cloud instances and reduce operational costs.
- Implemented digital signage across service desks and various campuses, to provide accessible information access.
- Authored technical documentation and procedural guides adopted across ITS operations.

EDUCATION

University of Minnesota School of Public Health, Minneapolis, MN

Master of Public Health - Public Health Data Science, GPA 3.73/4.00 *(Expected Jan 2026)*

- APEx — Pre-Statistics Modules for MPH Students
- ILE – Evaluation of US Federal Spending and Systems.

University of St. Thomas, St. Paul, MN

Bachelor of Science - Data Analytics (Biology Domain), Completed while employed full-time

- Minors - Biology (Biochemistry & Computational Biology courses), Applied Statistics.
- Capstone - Development of Obesity and Nutritional Trends Over Time in the United States Using NHANES.

AWARDS & RECOGNITION

- **ITS Outstanding Student Award**, University of St. Thomas (2023)
- **President's Student Leadership & Service Award Nominee**, University of Minnesota (2025)

LEADERSHIP AND VOLUNTEERING

Biostatistics Community Outreach & Engagement (BCOE), *Member (Sep 2023 - Present)*

School of Public Health Student Senate (SPHSS), *BSHD Representative (Oct 2023 - May 2025)*

- *Committees – Health Science Student Consultive Committee, Advocacy, Constitution Revision, Lounge Modernization.*

SKILLS

Languages: R, Python, SQL, Bash.

Tools: Shiny, APIs, Git, ArcGIS, Tableau, CI/CD, Microsoft Azure, Docker.

Competencies: Data Visualization, Data Communication, ETL Pipelines, Geospatial Visualization, Longitudinal Data Analysis, Machine Learning, Artificial Intelligence, Relational Databases, Columnar Data Architectures, System Design, Experimental Design, Consulting, Health Informatics, Electronic Health Records, HIPAA, PHI, SNOMED, ICD-10.

Statistical and Machine Learning Methods: Dimensionality Reduction, Supervised Learning, Unsupervised Learning, Regression, Regularization, Tree Models, Mixed Effects Models, Survival Analysis, Structural Equation Modeling, Natural Language Processing.

PROJECTS

Public Health Federal Funding Evaluation Platform & Dashboard (Grant Supported)

- Architected an in-house data warehousing platform supporting evaluation of U.S. federal public health funding.
- Integrated and standardized all HHS award records across grants, contracts, prime awards, and sub-awards.
- Applied normalization using pattern-matching (e.g. regex) and rules-based logic to standardize locations, government agencies, recipients, and award classifications.
- Implemented reconciliation logic to accurately link recipients, health departments, and related funding awards.
- Engineered a columnar storage architecture using Parquet and DuckDB, reducing data size by 75%+.
- Built an interactive Tableau dashboard to evaluate geographic distribution, funding trends, and program spending.
- Developed a GUI-based tool to manage data refreshes and support human-in-the-loop analysis.
  Methods & Tools: Python, Polars, SQL, DuckDB, Apache Parquet, Tableau (Hyper), Data Pipelines, Data Warehousing, Data Validation, Data Engineering, Data Visualization, Public Health Informatics.

Mapping Misery – Predicting Poor Physical and Mental Health Days in the U.S.

- Developed predictive models for poor physical and mental health days in 3,000+ U.S. counties using R.
- Engineered clusters based on 10 years of trends across 15 metrics using Gaussian Mixture Models (GMM) by transforming variables into 8 level categories with NAs for robust Multiple Correspondence Analysis (MCA).
- Achieved an R^2 of 0.844 (RMSE: 0.263) for physical health days with Elastic Net Regression.
  Methods & Tools: Data Pipelines, PCA, MCA, GMM Clustering, Feature Engineering, Longitudinal Data Analysis, Elastic Net Regression, Predictive Modeling, R, Geospatial Visualization, GIS, Data Visualization.

Tech Desk Tools Python

- Built a GUI-based desktop application for the University of St. Thomas Tech Desk to streamline common support tasks such as ticket creation, password resets, system diagnostics, and resource access in use continuously since February 2023.
- Redesigned and modernized legacy AutoHotKey application in Python, improving functionality and future scalability.
- Integrated role-specific admin features for senior student technicians based on assigned Azure roles.
- Deployed the application to all staff and student workers at the University of St. Thomas's Tech Desk.
  Methods & Tools: Python, PowerShell, Bash, Dashboards, Data Communication, GUI Development, Active Directory, Azure, Technical Writing, Packaging and Deployment.

Applied Biostatistics Introduction Modules (Grant Supported)

- Designed modular teaching materials introducing probability, inference, and regression through applied public health examples at the request of Public Health Administration & Policy program director
- Developed intuition-focused explanations to support students with apprehension toward statistics.
- Addressed recurring conceptual gaps identified by faculty and me from teaching.
- Emphasized ethical interpretation and justification of statistical choices in applied public health contexts.
  Methods & Tools: Biostatistics, Statistical Reasoning, Probability, Curriculum Design, Technical Writing, R, Data Communication, Public Health Education.

Data Inspection Dashboard | https://jonathanbarnes.shinyapps.io/DescriptiveApp/

- Developed an interactive tool that allows users to conduct baseline Exploratory Data Analyses (EDAs) without requiring R knowledge and accepts common dataset formats in public health.
- Automates and facilitates data cleaning, NA handling, imputation, and produces a data table with inline distribution plots and summary statistics.
  Methods & Tools: R, Shiny, Data Pipelines, Dashboards, Data Visualization, Data Communication, JavaScript.

Biostatistics Reference Guide

- Provides step-by-step instructions and R code for linear regression, logistic regression, Poisson regression, Cox proportional hazards models, survival analysis, ordinal regression, and log-binomial regression.
  Methods & Tools: R, Regression Analysis, Cox Proportional Hazard, Logistic Regression, Poisson Regression, Ordinal Regression, Log-binomial Regression, Tidyverse, Technical Writing.

Predicting Child Mortality and Cesarean Delivery Outcomes with Generalized Linear Mixed Models.

- Built Generalized Linear Mixed Models (GLMM) to investigate factors affecting child mortality and cesarean delivery.
- Determined a 1,124.5% increase in the odds of cesarean delivery for hospital births (95% CI: 618.7% - 1,986.3%, $p < 0.001$) — a result consistent with logical expectations.
  Skills & Tools: GLMM, GEE, R, Data Cleaning, Data Pipelines, Data Visualization, Feature Engineering, Collaboration.